

A uniform L^1 semigroup bound for homogeneous elliptic operators

Candidate full solution packet for Question 4.5 of arXiv:1805.08557

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Status. Candidate full solution, likely valid, awaiting specialist review. In the constant-coefficient setting of the source paper, Question 4.5 has an affirmative answer in the fastest listed decay regime, with $\beta = 0$. More precisely, the semigroup is uniformly bounded on $L^1(\mathbb{R}^d)$.

1 Source question and setting

Ben-Artzi and Einav consider $H = P(D)$ on $L^2(\mathbb{R}^d)$, where in the homogeneous elliptic subsection

$$P(\xi) = \sum_{|\alpha|=m} a_\alpha \xi^\alpha, \quad m \in \{2, 4, 6, \dots\},$$

has real coefficients and satisfies their Assumption A3. In particular, P is homogeneous of degree m , $P(0) = 0$, and

$$P(\xi) \geq c|\xi|^m \quad (\xi \in \mathbb{R}^d)$$

for some $c > 0$. The semigroup generated by $-P(D)$ is denoted here by $T_t = e^{-tP(D)}$; it is the source's P_t .

Question 4.5 asks whether for every such operator and every relevant initial datum u there are $C_2 = C_2(u) \geq 0$ and $\beta \in \mathbb{R}$ such that

$$\|T_t u\|_1^2 \leq \|u\|_1^2 + C_2 t^\beta \quad (t \geq 0).$$

An affirmative answer with $C_2 = 0$ or $\beta \leq 0$ gives the source's fastest listed decay rate, $O(t^{-d/m})$ [1].

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results in the literature for general operators as the ones we consider here. Based on the known results for the Laplacian and the fractional Laplacian one could ask:

Question 4.5. *Is it true that for every homogeneous elliptic operator of order m which satisfies Assumption A3 and which is the generator of a semigroup $(P_t)_{t \geq 0}$ there exist $C_2 = C_2(u) \geq 0$ and $\beta \in \mathbb{R}$ such that for every $t \geq 0$, $\|P_t u\|_{L^1}^2 \leq \|u\|_{L^1}^2 + C_2 t^\beta$?*

If the answer is 'yes', from Theorem 1.7 this conjecture leads to the following rate of convergence to equilibrium:

$$\text{Var}(P_t u) \leq \begin{cases} O((\log t)^{-\frac{d}{m}}) & \beta = \frac{d}{m}. \\ O(t^{\beta - \frac{d}{m}}) & 0 < \beta < \frac{d}{m}. \\ O(t^{-\frac{d}{m}}) & C_2 = 0 \text{ or } \beta \leq 0. \end{cases}$$

Figure 1: Question 4.5 and its stated consequence on page 14 of the source paper. The crop preserves the full rendered page width.

2 Proof intuition

Homogeneity makes every positive-time heat kernel a dilation of a single kernel. Indeed, if $k = \mathcal{F}^{-1}(e^{-P})$, then the kernel at time t is

$$k_t(x) = t^{-d/m} k(t^{-1/m}x).$$

Ellipticity forces e^{-P} to decay exponentially together with all of its derivatives, so k is a Schwartz function. Thus $k \in L^1$, while the dilation formula gives $\|k_t\|_1 = \|k\|_1$ for every $t > 0$. Young's inequality then yields a time-uniform L^1 operator norm. No positivity of the higher-order heat kernel is required.

3 The uniform kernel estimate

Lemma 1. *Let P be a real homogeneous polynomial of even degree m satisfying $P(\xi) \geq c|\xi|^m$. Set*

$$k = \mathcal{F}^{-1}(e^{-P}), \quad k_t = \mathcal{F}^{-1}(e^{-tP}) \quad (t > 0).$$

Then $k \in \mathcal{S}(\mathbb{R}^d)$,

$$k_t(x) = t^{-d/m} k(t^{-1/m}x), \quad \|k_t\|_{L^1} = \|k\|_{L^1}.$$

Proof. For every multi-index γ , repeated differentiation writes $\partial^\gamma(e^{-P})$ as a polynomial times e^{-P} . Since $P(\xi) \geq c|\xi|^m$, multiplication by any further monomial still leaves a bounded function. Hence $e^{-P} \in \mathcal{S}(\mathbb{R}^d)$. The Fourier transform is an automorphism of the Schwartz space, so $k \in \mathcal{S}(\mathbb{R}^d) \subset L^1(\mathbb{R}^d)$.

Homogeneity gives

$$e^{-tP(\xi)} = e^{-P(t^{1/m}\xi)}.$$

The Fourier scaling rule therefore yields $k_t(x) = t^{-d/m} k(t^{-1/m}x)$. A change of variables gives the asserted equality of L^1 -norms. $\square$

4 Full answer to Question 4.5

Theorem 2. *In the constant-coefficient homogeneous elliptic setting of Question 4.5, the semigroup $T_t = e^{-tP(D)}$ extends to a uniformly bounded convolution semigroup on $L^1(\mathbb{R}^d)$. Specifically, with*

$$M_P := \|\mathcal{F}^{-1}(e^{-P})\|_{L^1} < \infty,$$

one has

$$\|T_t u\|_{L^1} \leq M_P \|u\|_{L^1} \quad \text{for all } u \in L^1(\mathbb{R}^d), \quad t > 0.$$

Consequently Question 4.5 has an affirmative answer with

$$\beta = 0, \quad C_2(u) = \max\{M_P^2 - 1, 0\} \|u\|_{L^1}^2.$$

Proof. On Schwartz functions, Fourier diagonalization of the constant-coefficient operator gives

$$T_t u = \mathcal{F}^{-1}(e^{-tP} \widehat{u}) = k_t * u.$$

By Lemma 1 and Young's convolution inequality,

$$\|T_t u\|_1 \leq \|k_t\|_1 \|u\|_1 = M_P \|u\|_1.$$

This estimate extends the formula uniquely from Schwartz functions to all of L^1 , and on $L^1 \cap L^2$ it agrees with the original L^2 spectral semigroup. The multiplier identity gives the semigroup law first on Schwartz functions and then by density on L^1 . Moreover, $\int k = 1$, and the displayed dilations of k form a bounded approximate identity, so the extension is strongly continuous at $t = 0$.

Squaring the estimate and using the displayed choice of $C_2(u)$ proves

$$\|T_t u\|_1^2 \leq \|u\|_1^2 + C_2(u)t^0 \quad (t > 0).$$

At $t = 0$, $T_0 u = u$, so the source inequality also holds there (whether one defines $0^0 = 1$ or simply checks the endpoint separately). $\square$

Remark 3. If the semigroup is positivity preserving and mass preserving, then its convolution kernel has L^1 -norm one and the stronger contraction estimate $\|T_t u\|_1 \leq \|u\|_1$ holds. The proof above covers higher-order kernels that change sign, with a fixed constant M_P .

5 The resulting decay rate

The source proves the homogeneous Nash-type inequality

$$\|v\|_2^2 \leq C_P \mathcal{E}(v)^{d/(m+d)} \|v\|_1^{2m/(m+d)}, \quad \mathcal{E}(v) = \|P(D)^{1/2} v\|_2^2.$$

For completeness, the uniform L^1 estimate also gives the stated decay directly. For $u \in L^1 \cap L^2$, put $y(t) = \|T_t u\|_2^2$. The spectral calculus gives $y'(t) = -2\mathcal{E}(T_t u)$. Rearranging the Nash inequality and using Theorem 2 yields

$$y'(t) \leq -c_P (M_P \|u\|_1)^{-2m/d} y(t)^{1+m/d}.$$

Integration gives

$$y(t) \leq \left(y(0)^{-m/d} + c'_P (M_P \|u\|_1)^{-2m/d} t \right)^{-d/m} = O(t^{-d/m}).$$

Thus the answer realizes the $\beta \leq 0$ line predicted immediately after Question 4.5.

6 Verification, scope, and novelty check

The proof uses only four auditable facts: the source’s operator is a Fourier multiplier; ellipticity makes e^{-P} Schwartz; homogeneity gives exact kernel scaling; and Young’s inequality converts this into an L^1 bound. The quantifiers are stronger than requested: M_P is independent of both t and u , while the allowed additive constant is then chosen for each u . The argument does not assume positivity, and it handles $t = 0$ separately.

The conclusion is scoped to the source’s Section 4 setting: scalar, constant-coefficient operators $P(D)$ on $\mathbb{R}^d$, with a real positive homogeneous polynomial symbol. It does not claim the same bound for variable-coefficient elliptic operators or for arbitrary nonhomogeneous symbols.

A bounded literature check through 26 August 2026 searched the run’s registry, solution, attempt, and proof-gap indexes; arXiv:1805.08557 and its exact Question 4.5 wording; and primary-source searches combining “homogeneous elliptic operator”, “heat kernel”, “ L^1 ”, and “semigroup”. It found the source paper and general heat-kernel literature, but no later paper explicitly identifying itself as an answer to Question 4.5. The kernel-scaling ingredients are classical, so this search does not establish historical priority. The packet’s claim is the direct identification and complete proof of the answer to the source question.

The highest-value specialist check is the interpretation of “homogeneous elliptic operator” exactly as fixed by the surrounding constant-coefficient subsection. Under that stated context, the proof is complete.

References

- [1] J. Ben-Artzi and A. Einav, *Weak Poincaré inequalities in the absence of spectral gaps*, Annales Henri Poincaré 21 (2020), 359–375; arXiv:1805.08557v5.
- [2] L. Grafakos, *Classical Fourier Analysis*, 3rd ed., Graduate Texts in Mathematics 249, Springer, 2014.